# University of Barishal

## Department of Computer Science and Engineering

### Final Exam - Semester: 6th

**Course:** Course Title: Simulation and Modeling (CSE-3209) **Session:** 20-21

**Time:** 180 minutes | **Marks:** 60.00 **Teacher:** Dr. Rahat Hossain Faisal

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*Answer any five Questions from the followings.*

**1.**

**a)** How simulation plays important role in new innovation?

**b)** In which situations simulation is not a better choice?

**c)** What are different methods of studying a system before designing a simulation?

**d)** Classify simulation models.

**2.**

**a)** Briefly explain the differences between discrete system and continuous system with suitable figures.

**b)** Discuss the concept of "Time Advance Mechanism" with an example.

**c)** What are different components of a Discrete-event simulation models? Explain.

**3.**

**a)** Write short notes on the followings including their scope of implementation in simulation process: i) Exponential Distribution ii) Triangular Distribution iii) Lognormal Distribution iv) Poisson Process v) Weibull Distribution vi) Geometric Distribution

**4.**

**a)** Define the following queuing system characteristics: (i) calling population (ii) Arrival process.

**b)** A classical inventory problem concerns the purchase and sale of newspapers. The paper seller buys the papers for 33 cents each and sells them for 50 cents each. (The lost profit from excess demand is 17 cents for each paper demanded that could not be provided.) Newspapers not sold at the end of the day are sold as scrap for 5 cents each. (the salvage value of scrap papers) . There are three types of Newsday's, "good," "fair," and "poor," with probabilities of 0.35, 0.45, and 0.20 , respectively. The problem is to determine the optimal number of papers the newspaper seller should purchase. This will be accomplished by simulating demands for 20 days and recording profits from sales each day. The demand table and distribution of type of Newsday are given below. Simulation Task Simulate for the purchase of 70 newspapers per day for 20 days and compute the profit of the news dealer. Random Numbers for Simulation Random numbers for Type of News day: 94, 77, 49, 45, 43, 32, 49, 00, 16, 24, 31, 14, 41, 61, 85, 08, 15, 97, 52, 78 Random numbers for Demand: 80, 20, 15, 88, 98, 65, 86, 73, 24, 60, 60, 29, 18, 90, 93, 73, 21, 45, 76, 96

**5.**

**a)** What are the problems may occur during the generation of random numbers?

**b)** Define linear congruential method for generating random number. Use the method with X 0 =27, a=17, c=43, m=100 to generate 4 numbers

**c)** What are the methods used for testing a random number? Test uniformity for 0.44, 0.81, 0.14, 0.05, 0.93 using Kolmogorov-Smirnov test (α=0.05, D α =0.565).

**6.**

**a)** What is the Kendall notation of Queuing System?

**b)** Define a Markov chains and its application.

**c)** Use the mixed congruential method to generate sequence of three two-digit random numbers with X 0 =37, a=7, c=29, m=100.

**7.**

**a)** Explain how simulation and modelling can play an important role in Manufacturing and Material Handling System.

**b)** Briefly describe probable simulation processes in a Manufacturing System [Use an appropriate example].

**c)** Define verification in simulation process? Describe techniques to perform verification on simulation model.

**8.**

**a)** Monte Carlo simulation is a special case of stochastic simulation? Comment.

**b)** What method did we use to develop the model for the ATM (bank) case study?

**c)** What is SIMSCRIPT? What is its importance? Discuss its various functions.

**d)** Write short note on Statistical output analysis.